

Assignment 4

1 Attention Exploration (14 points)

- (a) i. The categorical distribution α puts almost all of its weight on α_j when the score $k_j^\top q$ is much larger than every other score $k_i^\top q$; equivalently, $\sum_{i \neq j} \exp(k_i^\top q - k_j^\top q) \ll 1$.
- ii. Under this condition, $\alpha_j \approx 1$ and $\alpha_i \approx 0$ for every $i \neq j$. Therefore, $c = \sum_{i=1}^n \alpha_i v_i \approx v_j$.

- (b) Let $\ell > 0$ be large and choose

$$q = \ell(k_a + k_b).$$

Since the keys are orthogonal unit vectors, $k_a^\top q = k_b^\top q = \ell$, whereas $k_i^\top q = 0$ for $i \notin \{a, b\}$. Hence

$$\alpha_a = \alpha_b = \frac{e^\ell}{2e^\ell + (n-2)} = \frac{1}{2 + (n-2)e^{-\ell}} \approx \frac{1}{2}$$

whenever $e^\ell \gg n-2$. The remaining attention weights are negligible, so $c \approx \frac{1}{2}(v_a + v_b)$.

- (c) i. To distinguish the covariance scale from the attention weights α_i , let ε denote the scalar called α in the question. Thus, $\Sigma_i = \varepsilon I$ with $\varepsilon \rightarrow 0$. Choose

$$q = \ell(\mu_a + \mu_b), \quad e^\ell \gg n-2.$$

Write $k_i = \mu_i + \delta_i$, where $\delta_i \sim \mathcal{N}(0, \varepsilon I)$. Since the means are orthogonal unit vectors,

$$k_a^\top q \approx k_b^\top q \approx \ell, \quad k_i^\top q \approx 0 \quad (i \notin \{a, b\}).$$

Therefore, $\alpha_a \approx \alpha_b \approx \frac{1}{2}$ and $c \approx \frac{1}{2}(v_a + v_b)$. To make the conditions on ℓ precise, first note that the random score perturbation $\delta_i^\top q$ is a linear projection of a Gaussian vector and is therefore Gaussian. Its mean and variance are

$$\mathbb{E}[\delta_i^\top q] = 0, \quad \text{Var}(\delta_i^\top q) = q^\top (\varepsilon I) q = \varepsilon \|q\|^2 = 2\varepsilon \ell^2,$$

where $\|q\|^2 = \ell^2 \|\mu_a + \mu_b\|^2 = 2\ell^2$. Hence

$$\delta_i^\top q \sim \mathcal{N}(0, 2\varepsilon \ell^2),$$

whose standard deviation is $\sqrt{2}\ell\sqrt{\varepsilon}$. We may thus choose

$$\log(n-2) \ll \ell \ll \varepsilon^{-1/2}.$$

The lower bound implies $e^\ell \gg n-2$, suppressing the total contribution of the irrelevant keys. The upper bound is equivalent to $\ell\sqrt{\varepsilon} \ll 1$, ensuring that the random score perturbations are small and that the scores of k_a and k_b remain approximately equal.

- ii. Now $\Sigma_a = \varepsilon I + \frac{1}{2}\mu_a\mu_a^\top$. Ignoring the vanishing orthogonal noise, write $k_a \approx r\mu_a$, where the random magnitude

$$r = \mu_a^\top k_a \sim \mathcal{N}\left(1, \varepsilon + \frac{1}{2}\right).$$

Using the query from part (i), the two relevant scores satisfy

$$q^\top k_a \approx \ell r, \quad q^\top k_b \approx \ell, \quad \frac{\alpha_a}{\alpha_b} \approx e^{\ell(r-1)}.$$

Thus, the softmax exponentially amplifies fluctuations in r . When $r > 1$, c is close to v_a ; when $r < 1$, c is close to v_b ; only when r is very close to 1 is c close to $\frac{1}{2}(v_a + v_b)$. Consequently, individual samples produce outputs that vary substantially between vectors near v_a and v_b , so the variance of c is much larger than in part (i), even though its average over the symmetric fluctuations may remain near $\frac{1}{2}(v_a + v_b)$.

- (d) i. Let $\ell > 0$ be large and choose two different queries

$$q_1 = \ell\mu_a, \quad q_2 = \ell\mu_b.$$

When $\Sigma_i = \varepsilon I$ and $\varepsilon \rightarrow 0$, we have $k_i \approx \mu_i$. Thus, for q_1 , the score of k_a is approximately ℓ , while all other scores are approximately zero; similarly, q_2 selects k_b . If $e^\ell \gg n - 1$ (and $\ell\sqrt{\varepsilon} \ll 1$), then

$$c_1 \approx v_a, \quad c_2 \approx v_b, \quad c = \frac{1}{2}(c_1 + c_2) \approx \frac{1}{2}(v_a + v_b).$$

- ii. With $\Sigma_a = \varepsilon I + \frac{1}{2}\mu_a\mu_a^\top$, write $k_a \approx r\mu_a$, where $r \sim \mathcal{N}(1, \varepsilon + \frac{1}{2})$. The scores relevant to the first head are

$$q_1^\top k_a \approx \ell r, \quad q_1^\top k_i \approx 0 \quad (i \neq a).$$

Ignoring the cases $q_1^\top k_a < 0$ as permitted, the first head still selects v_a whenever $\ell r \gg \log(n - 1)$. Hence $c_1 \approx v_a$, although it has some residual variance when r is positive but close to zero. For the second head,

$$q_2^\top k_b \approx \ell, \quad q_2^\top k_a \approx 0,$$

because $\mu_a^\top \mu_b = 0$. Therefore, $c_2 \approx v_b$ and has vanishingly small variance. Consequently,

$$c = \frac{1}{2}(c_1 + c_2) \approx \frac{1}{2}(v_a + v_b)$$

across samples. In particular, since c_2 is nearly deterministic,

$$\text{Var}(c) \approx \frac{1}{4} \text{Var}(c_1),$$

so the output is substantially more stable than the single-headed construction in part (c).

- (e) Multi-headed attention assigns v_a and v_b to separate heads, so a change in the magnitude of k_a affects mainly the head responsible for v_a rather than changing the competition between v_a and v_b within one softmax. Averaging the head outputs therefore preserves both values more reliably and reduces the variance of the final output.